

8. Agent Assistants

Rice Business 2026

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From a Tool to a Teammate

A research team

- Many agents, different roles, working together
- They propose, attack, verify, and re-check each other

A personal assistant

- One agent that carries work across sessions
- Remembers context, uses your tools, acts on your behalf

Two shapes where agents pay off: a *panel* that pressure-tests ideas and data, and an *assistant* that just gets recurring things done.

Not the Same as Day 6

Day 6 — Student Projects

- Agents students **build** as the deliverable
- The product is the point; the learning is the design and the verification

Today — Your Assistants

- Agents that **work for you**, on your research and your tasks
- The product is *your* output; the agent is the labor

Same underlying idea — an agent that decides and acts — pointed two ways: something students make, versus something that makes your week lighter.

A Research Team

Many Roles, One Question

Debate

- A proposer generates ideas; an adversary attacks them
- Disagreement surfaces the weak points early

Ground it

- A verifier checks claims against your own library
- An analyst builds the data and runs the empirics

Check it

- A replicator re-derives the data build *and* the result, independently
- Two paths to the same number, or a red flag

The coauthor skill orchestrates this: you are the coordinator; a debate panel and tool-using subagents do the work in a local folder — no repo, nothing published without you.

Why a Team Beats One Agent

A single agent that both proposes *and* checks its own work grades its own homework. Splitting the roles — **propose, attack, verify, replicate** — is how you catch the beautiful-but-wrong result before it reaches a referee.

Personal Assistants

An Agent That Carries the Work

Give it a goal, not a script

- You describe the outcome; it plans the steps
- It uses tools — files, code, the web — to act
- State persists, so today builds on yesterday

Good first jobs

- A weekly digest of new papers in your area
- Reformat and check a dataset on arrival
- Turn meeting notes into tracked action items

The recurring, rule-ish tasks that eat an hour a week are the sweet spot — worth automating, low cost if a run is imperfect.

What Makes It an Agent

Not just answering

- It decides *what to do next*, then does it
- It checks its own output and retries
- It stops and asks when it is stuck

Kept in bounds

- You choose which tools it may use
- You review before anything irreversible
- Give it the reach the task needs — and no more

The skill you are building is not prompting — it is describing the goal well and judging the result. Delegation, not dictation.

Why This Belongs at the End

Everything until now was you **directing a tool**. An agent is you **delegating a task** — a research team that pressure-tests your work, or an assistant that clears the recurring load. The new skill is knowing what to hand off, and how to check what comes back.

Breakout

Questions to Discuss

- What recurring task would you hand to a standing assistant first?
- For a question in your field, what roles would your ideal agent team have?
- What tools would you *withhold* from an agent working on real data?
- Where is the line between useful delegation and losing track of what was done?